

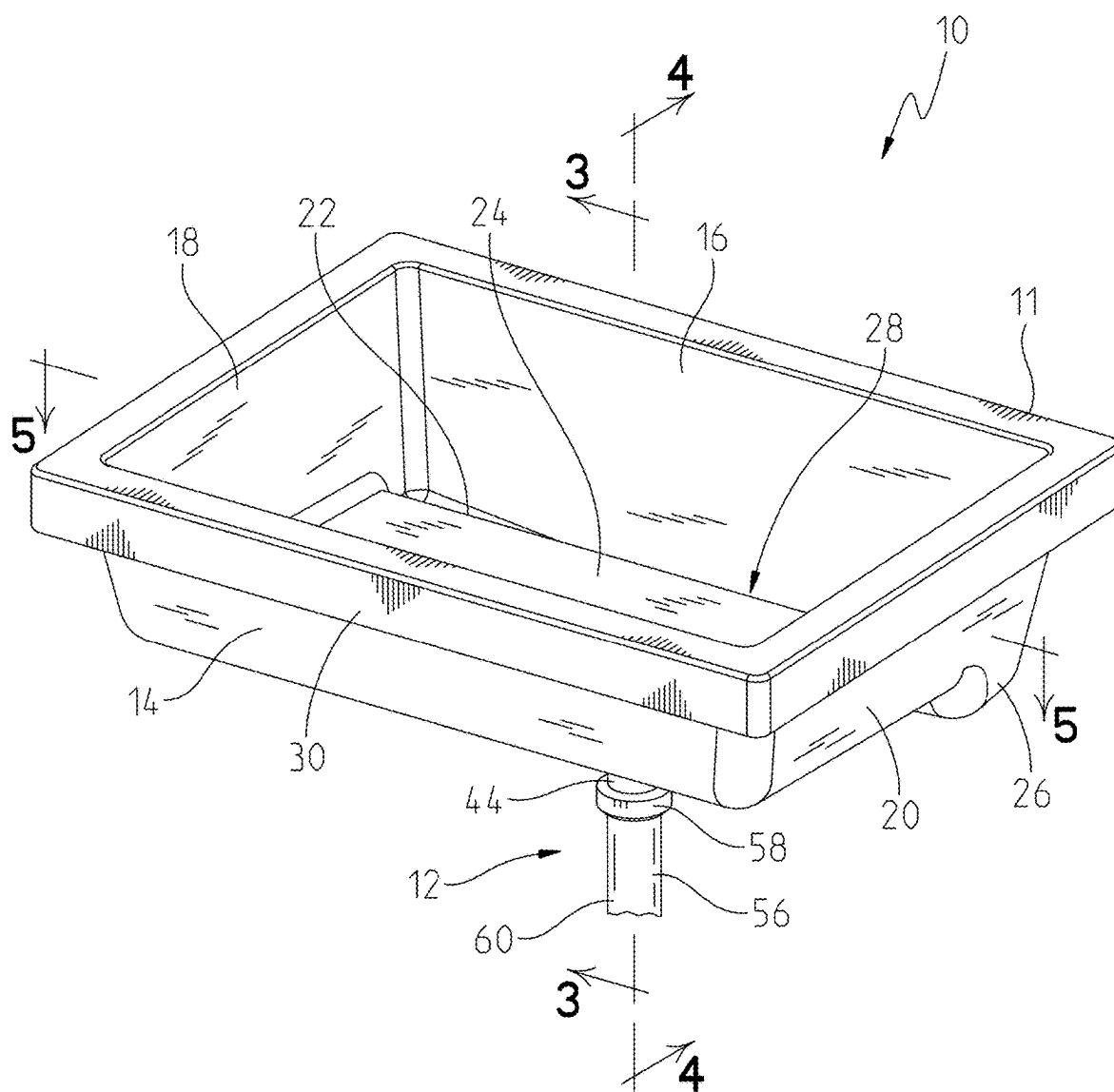


Fig. 1

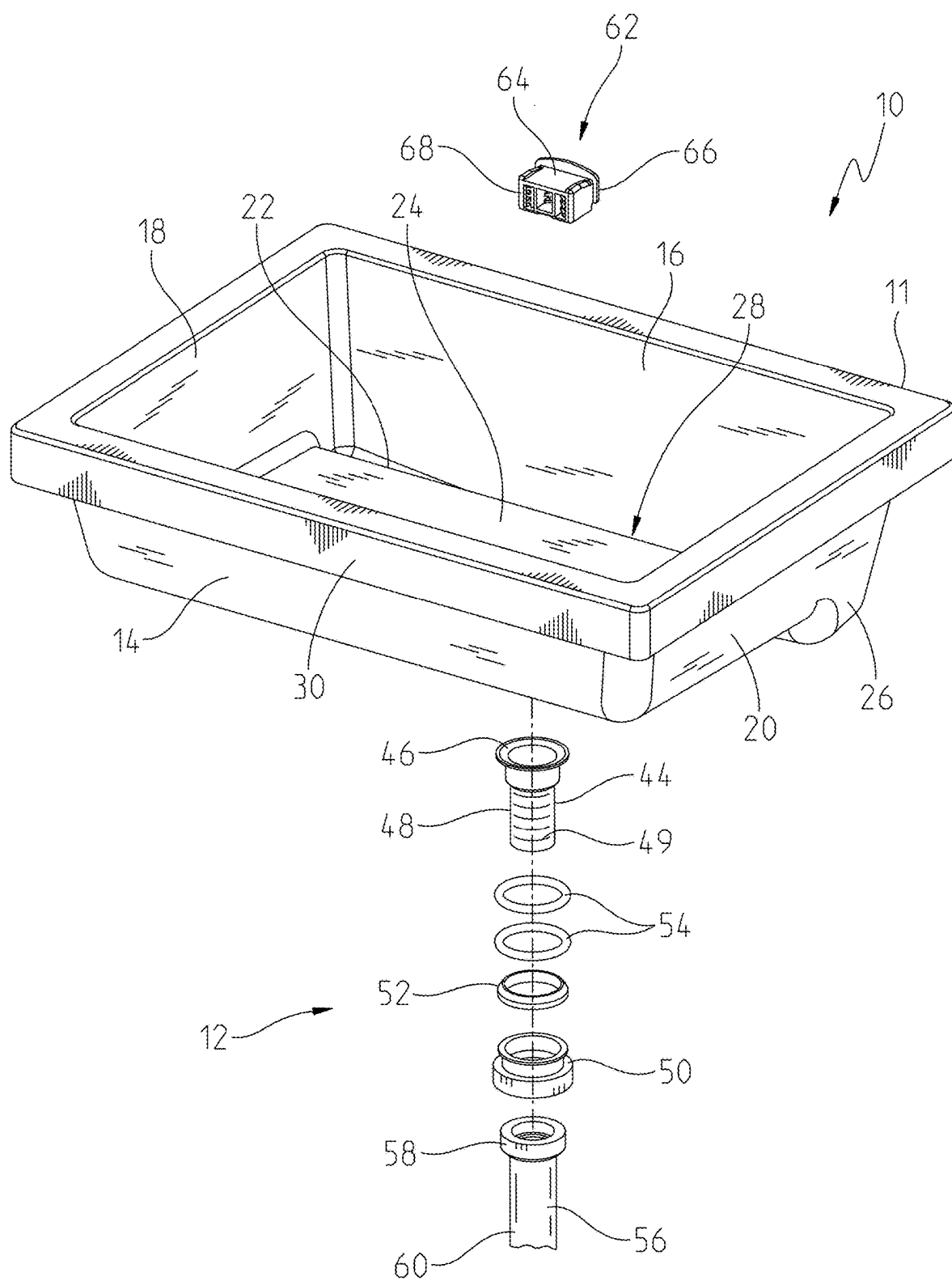


Fig. 2

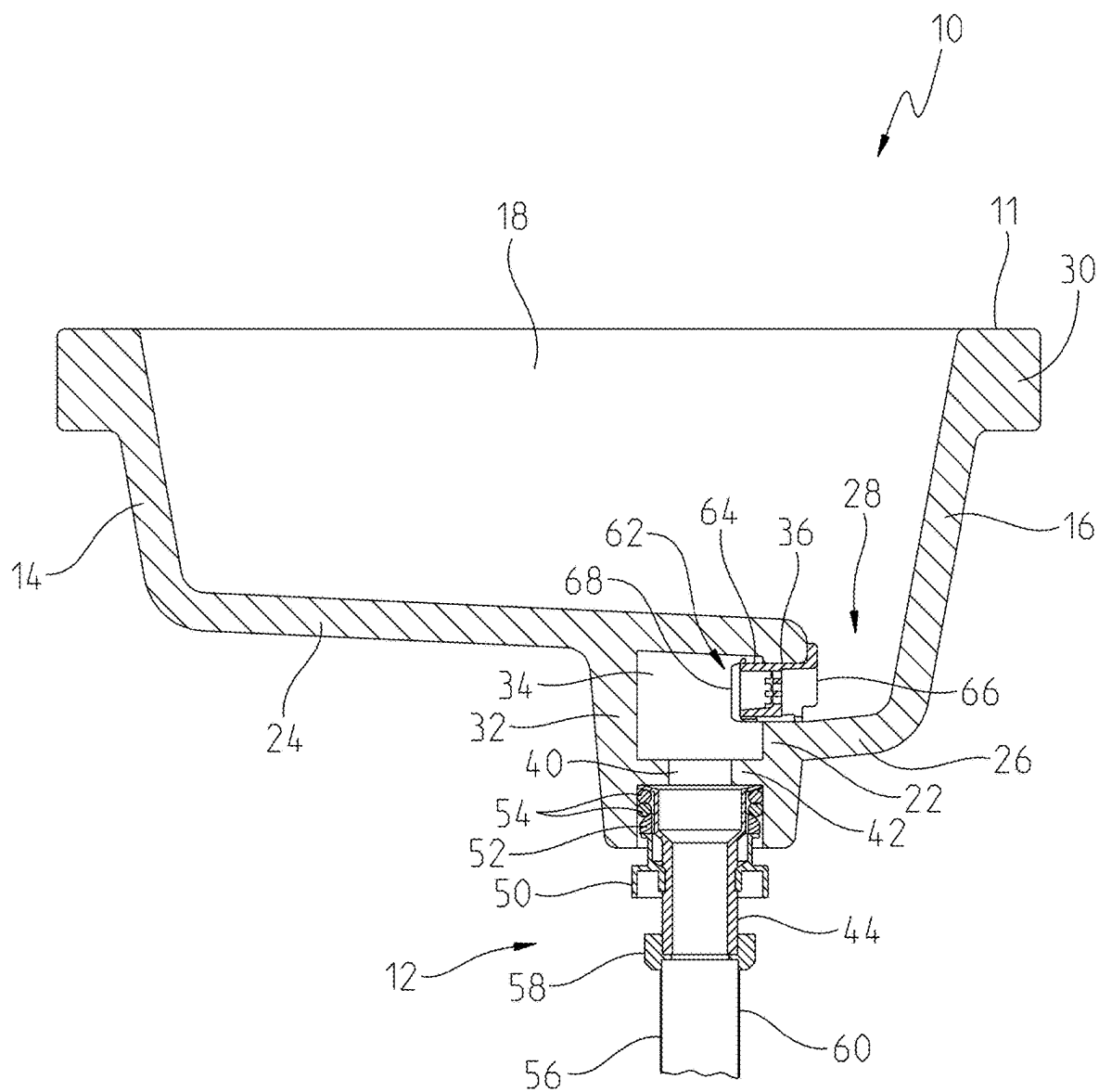


Fig. 3

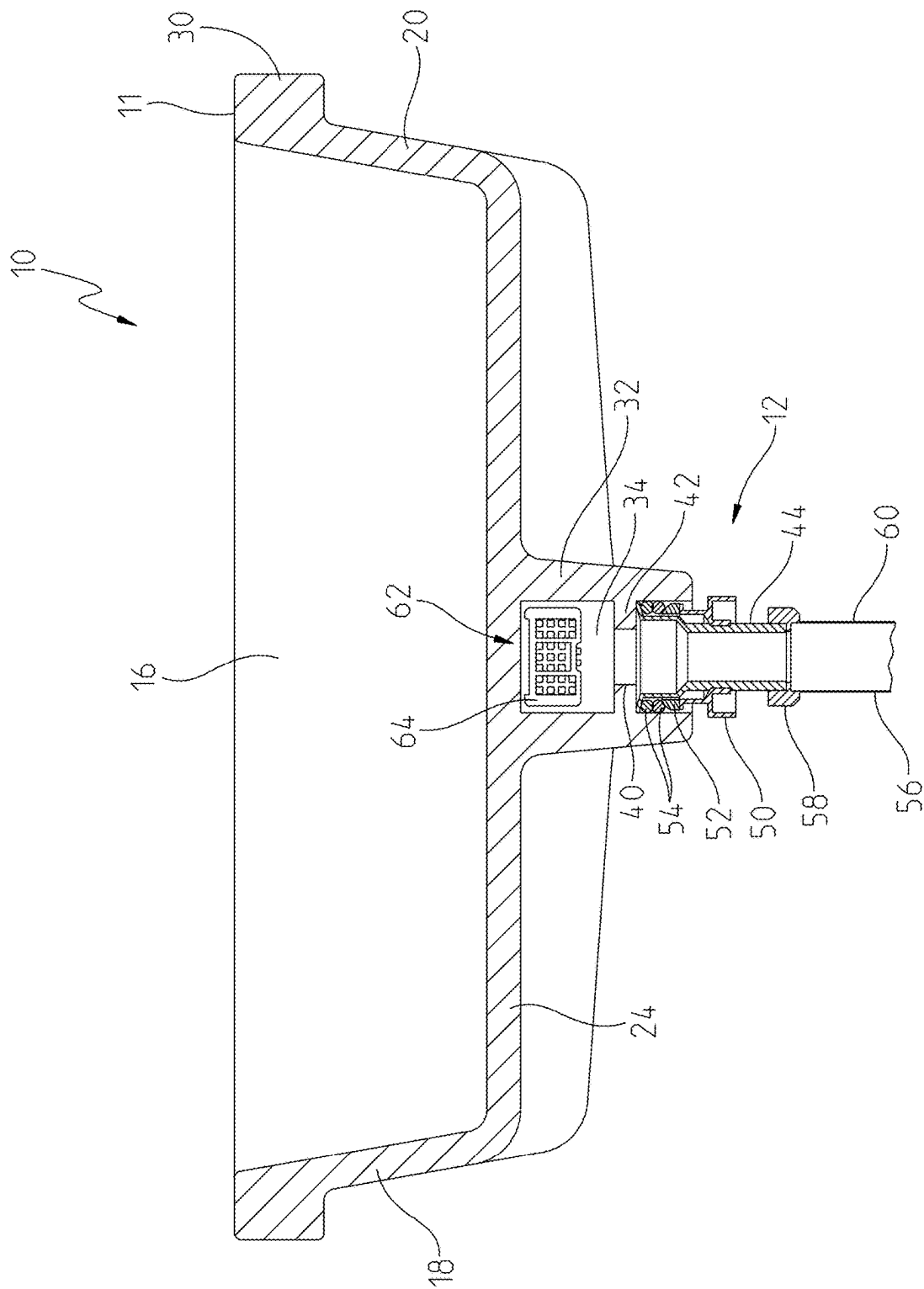


Fig. 4

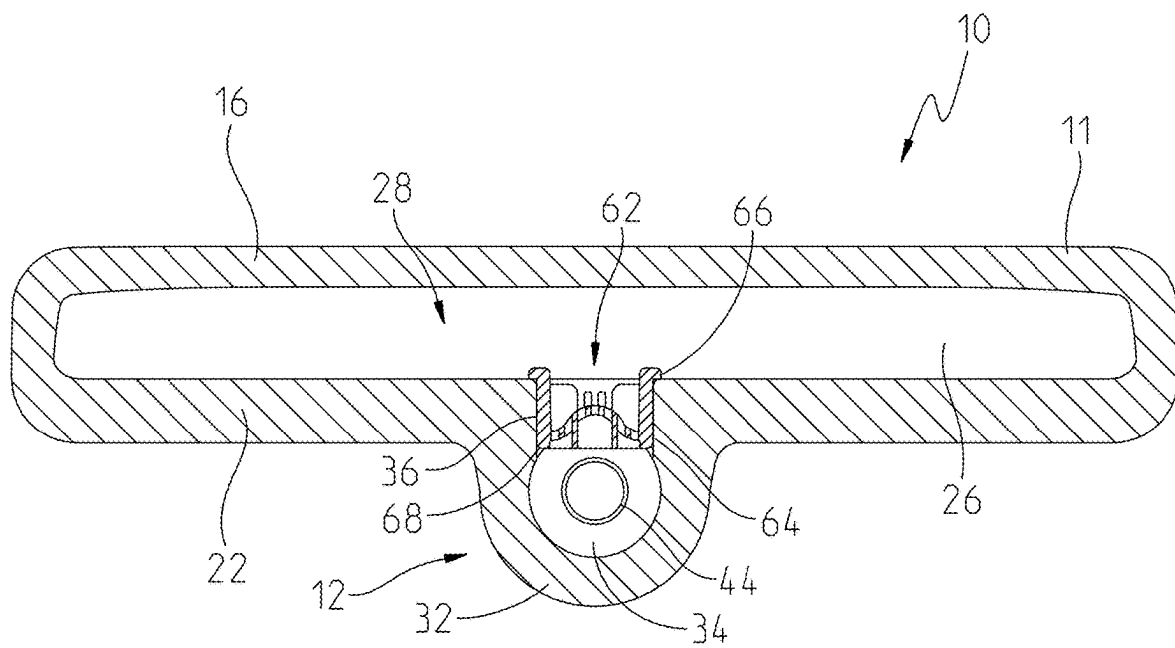


Fig. 5

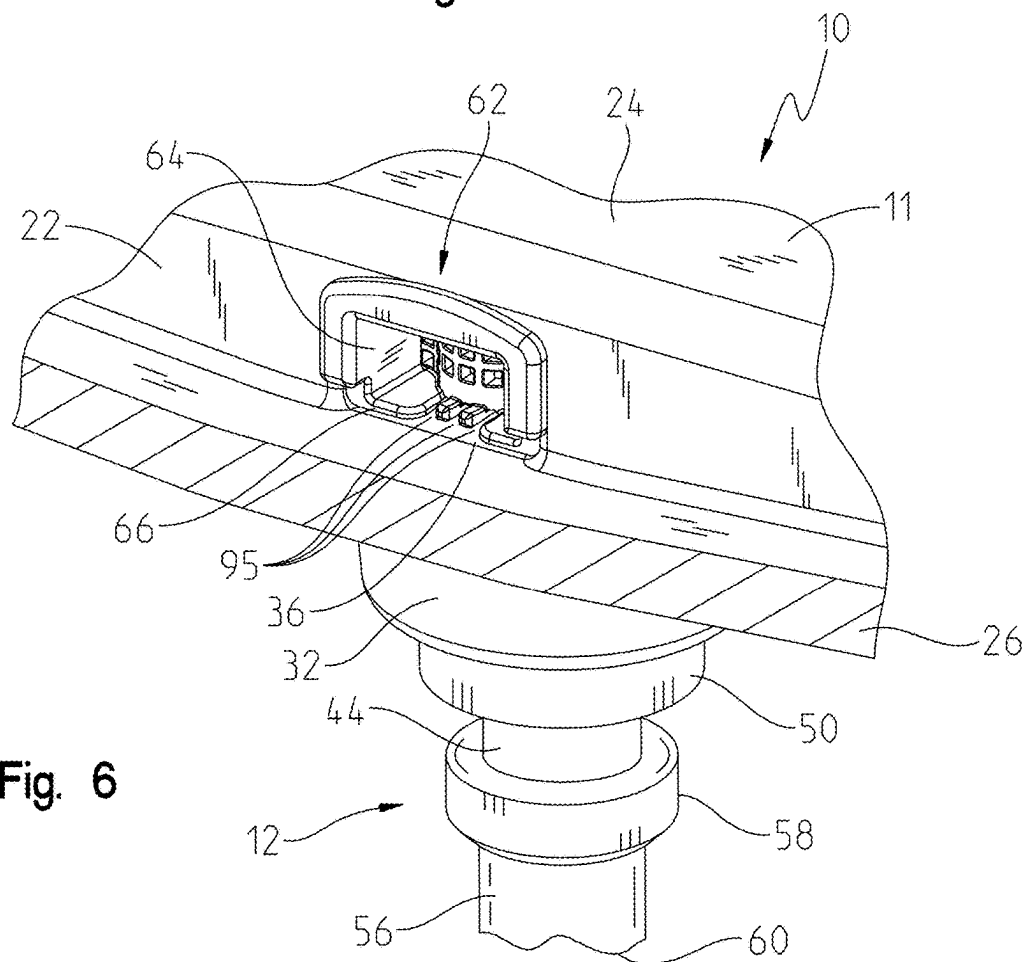


Fig. 6

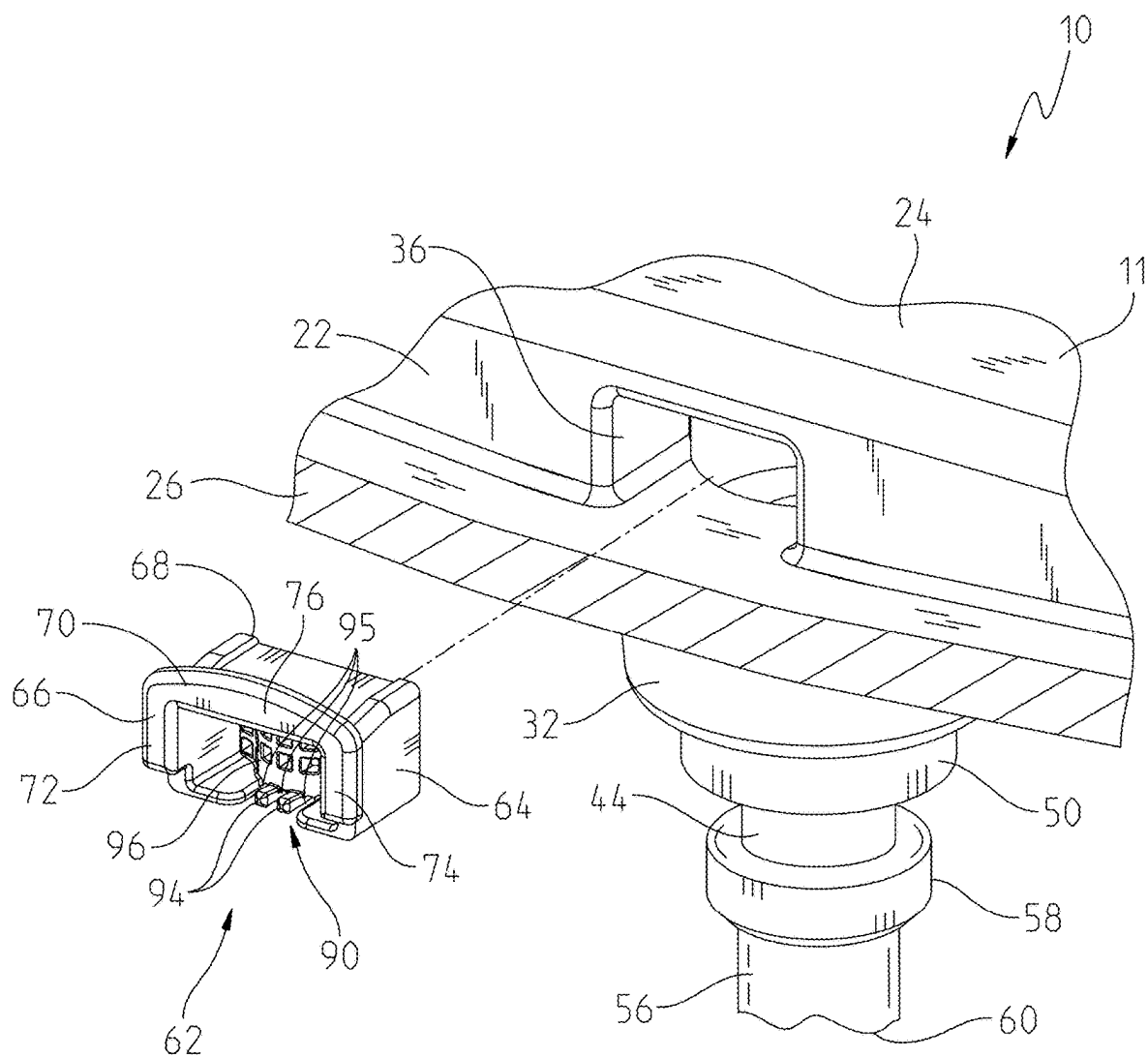


Fig. 7

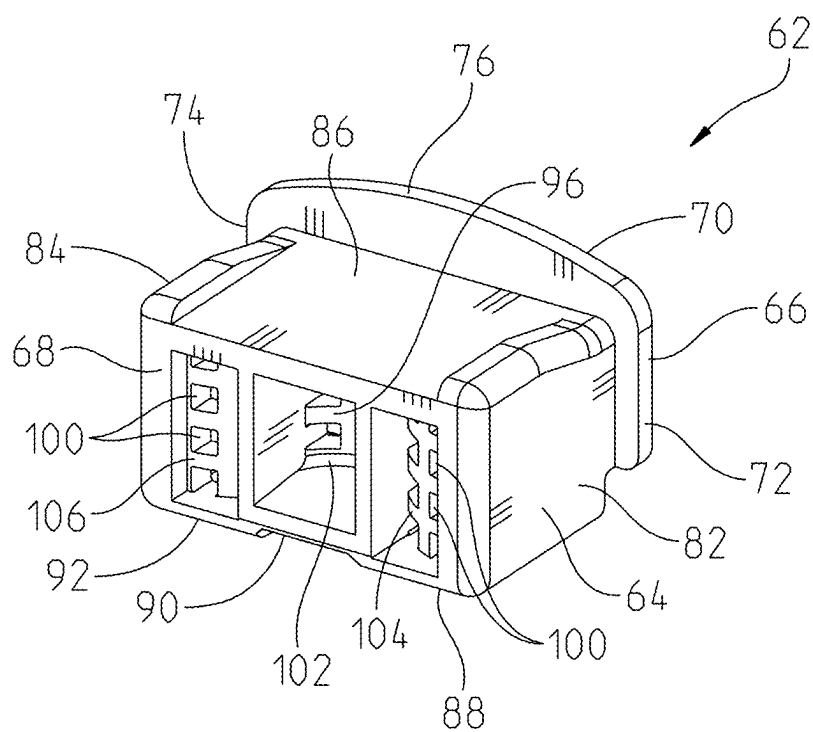


Fig. 8

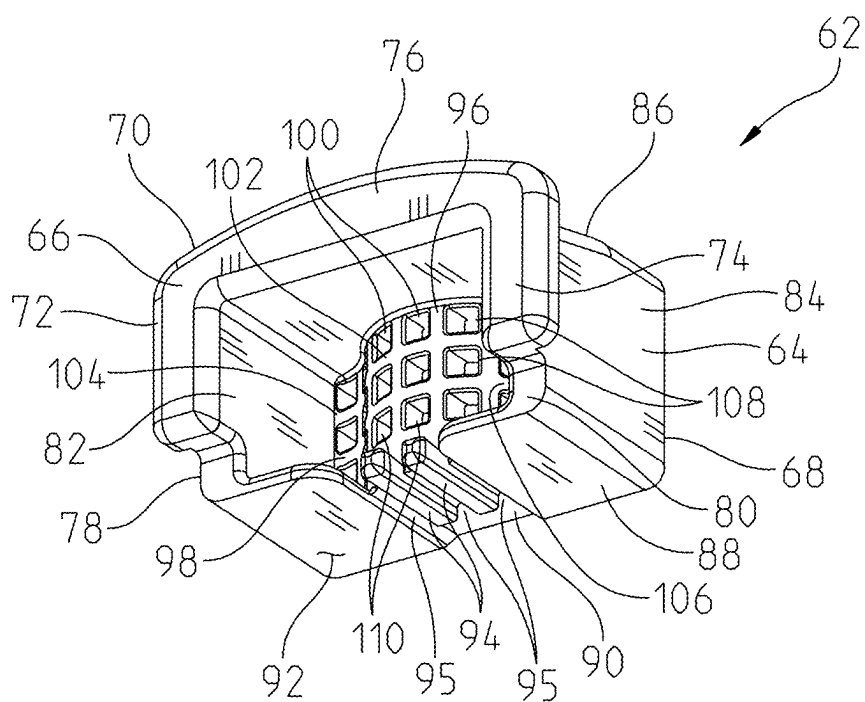


Fig. 9

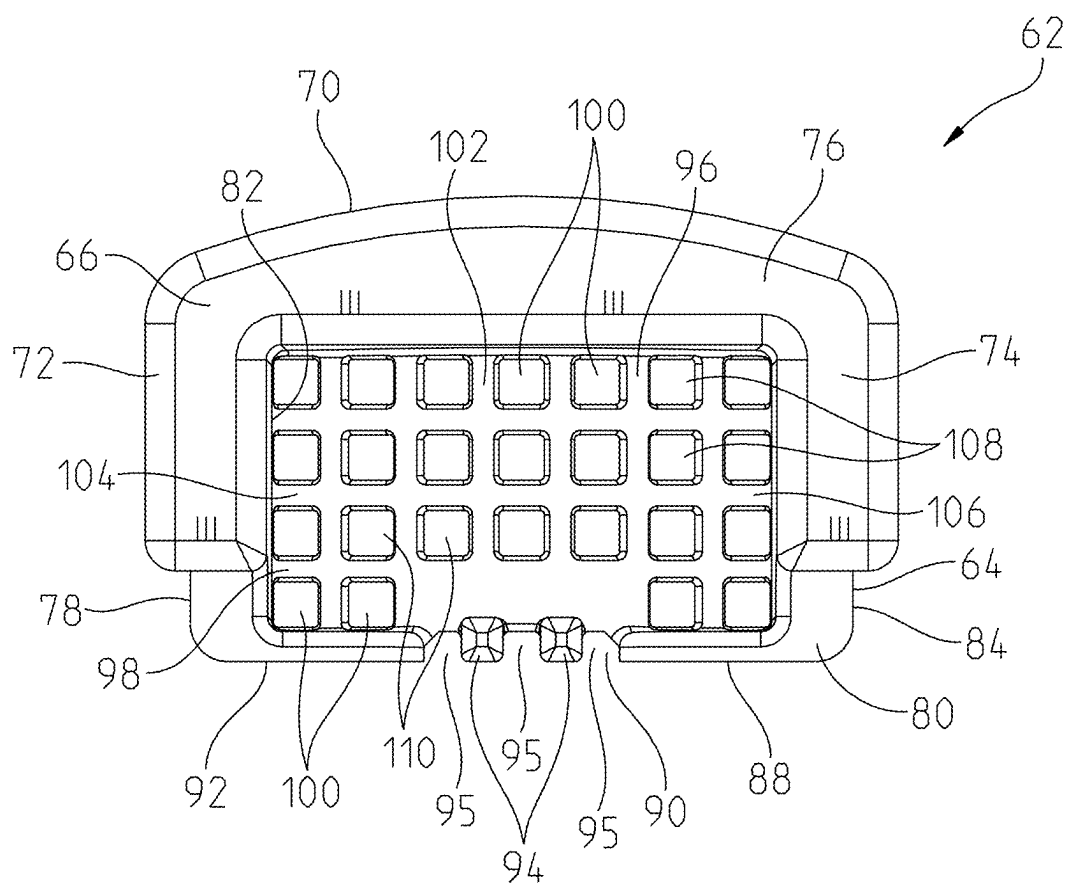


Fig. 10

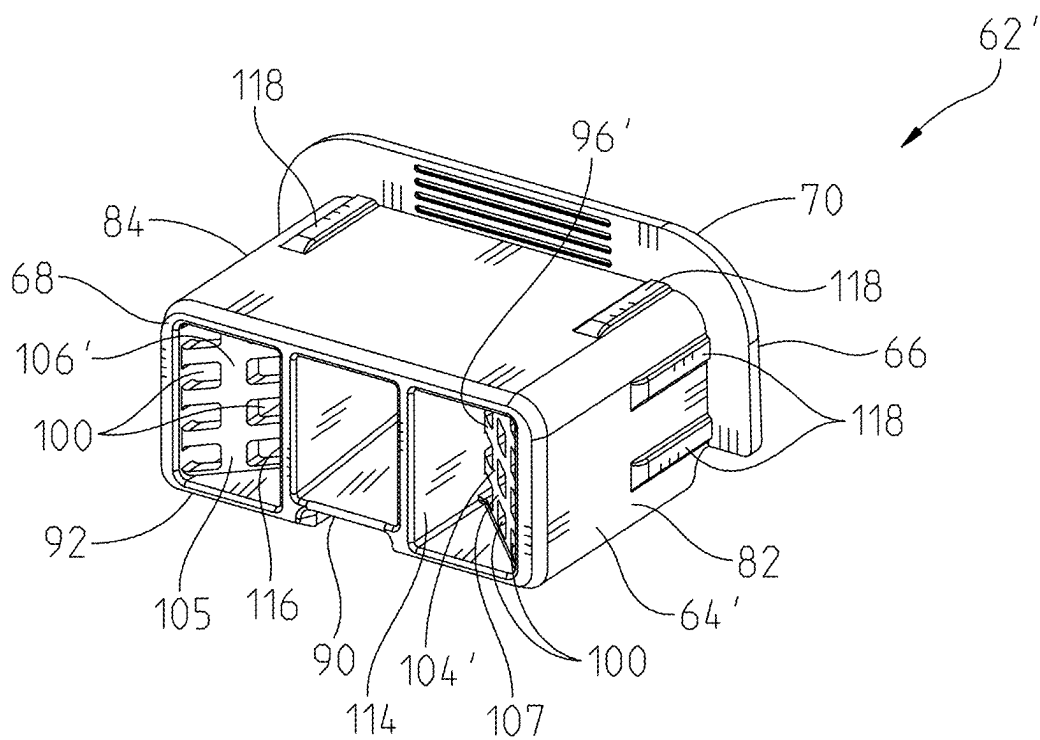


Fig. 11

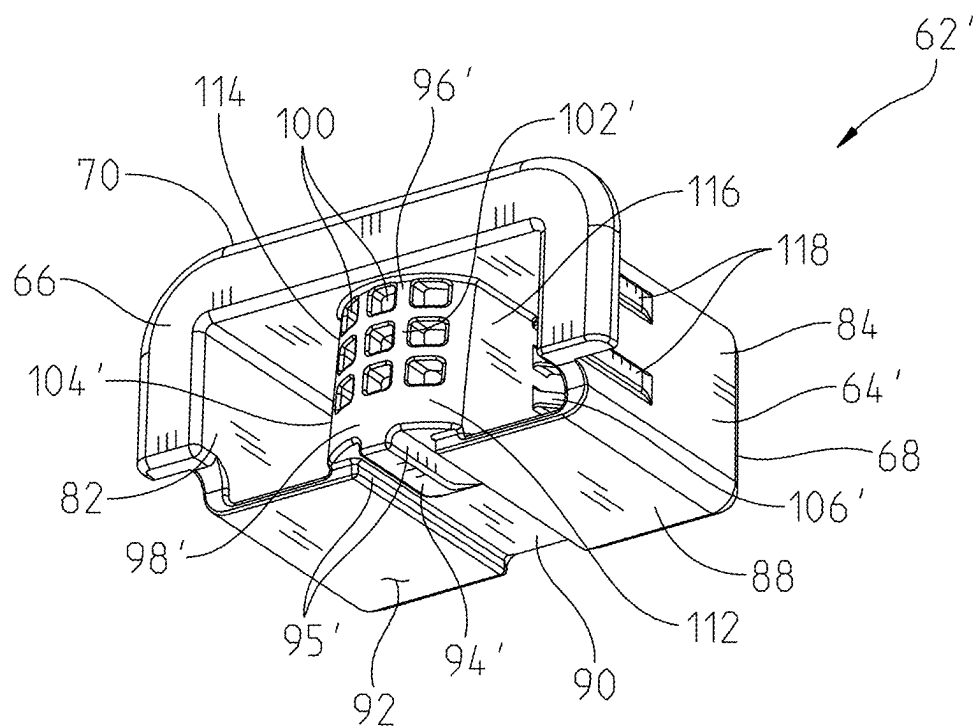


Fig. 12

SCREEN FOR A HIDDEN DRAIN SINK

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to U.S. Provisional Patent Application Ser. No. 63/556,649, filed Feb. 22, 2024, the disclosure of which is expressly incorporated herein by reference.

BACKGROUND AND SUMMARY OF THE DISCLOSURE

[0002] The present invention relates generally to drain assemblies for sink basins and, more particularly, to a drain assembly including a screen removably coupled to a hidden drain sink.

[0003] Hidden drain sinks are known to include a drain assembly installed from underneath a sink basin and which cannot be easily accessed once installed. Given the location of the drain in current hidden drain sinks, the end user may find it difficult to remove and/or prevent clogs.

[0004] An illustrative drain screen of the present disclosure is removably supported within the sink basin. More particularly, the user can easily remove the drain screen, should it become blocked with hair and/or debris, clean it and then replace it.

[0005] The illustrative drain screen is assembled from above and to the front of the sink basin rather than from the underside so that users can clean the drain without requiring disassembly of the drain outlet. Illustratively, the drain screen is made of a flexible material, such as silicone, that makes it easier for removal and cleaning.

[0006] According to an illustrative embodiment of the present disclosure, a drain screen for a hidden drain sink includes a resilient body extending axially between a proximal end and a distal end. A mounting flange is supported by the proximal end of the body. A strainer including a wall is supported by the body distally of the flange. The strainer includes a wall having a plurality of openings, the wall being curved axially towards the proximal end of the body.

[0007] According to a further illustrative embodiment of the present disclosure, a drain screen for a hidden drain sink includes a resilient body extending axially between a proximal end and a distal end. A mounting flange is supported by the proximal end of the body. A strainer is supported by the body distally of the flange. The strainer includes a wall having a plurality of openings, the wall including a center section positioned intermediate a first side section and a second side section. The center section extends proximally of the first side section and the second side section.

[0008] According to another illustrative embodiment of the present disclosure, a hidden drain sink includes a basin having a first side wall, and a second side wall laterally spaced from the first sidewall. A front wall extends between the first side wall and the second side wall. A rear wall extends between the first side wall and the second side wall, and is longitudinally spaced from the front wall. An intermediate wall is positioned between the front wall and the rear wall. A drain opening is defined within the intermediate wall. A drain channel extends between the intermediate wall and the rear wall. A bottom wall extends between the first side wall and the second side wall. A drain screen includes a resilient body extending axially between a proximal end and a distal end. The resilient body of the drain screen is

received within the opening of the intermediate wall of the basin. A strainer is supported by the body distally of the flange. The strainer includes a wall having a plurality of openings.

[0009] Additional features and advantages of the present invention will become apparent to those skilled in the art upon consideration of the following detailed description of the illustrative embodiment exemplifying the best mode of carrying out the invention as presently perceived.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] A detailed description of the drawings particularly refers to the accompanying figures, in which:

[0011] FIG. 1 is a perspective view of an illustrative drain assembly coupled to a sink basin of a hidden drain sink;

[0012] FIG. 2 is an exploded perspective view of the illustrative drain assembly of FIG. 1;

[0013] FIG. 3 is a cross-sectional view of the illustrative drain assembly, taken along line 3-3 of FIG. 1;

[0014] FIG. 4 is a cross-sectional view of the illustrative drain assembly, taken along line 4-4 of FIG. 1;

[0015] FIG. 5 is a cross-sectional view of the illustrative drain assembly, taken along line 5-5 of FIG. 1;

[0016] FIG. 6 is a detailed perspective view of the illustrative drain assembly of FIG. 1, with a partial cut-away of the sink basin;

[0017] FIG. 7 is a detailed perspective view of the illustrative drain assembly similar to FIG. 6, showing the screen removed from the sink basin;

[0018] FIG. 8 is a front perspective view of an illustrative screen of the drain assembly of FIG. 1;

[0019] FIG. 9 is a rear perspective view of the illustrative screen of FIG. 8;

[0020] FIG. 10 is a rear plan view of the illustrative screen of FIG. 8;

[0021] FIG. 11 is a front perspective view of another illustrative screen; and

[0022] FIG. 12 is a rear perspective view of the illustrative screen of FIG. 11.

DETAILED DESCRIPTION OF THE DRAWINGS

[0023] The embodiments of the invention described herein are not intended to be exhaustive or to limit the invention to the precise form disclosed. Rather, the embodiments selected for the description have been chosen to enable one skilled in the art to practice the invention.

[0024] Referring initially to FIGS. 1 and 2, a hidden drain sink 10 includes a sink basin 11 supporting a drain assembly 12. The sink basin 11 includes a front wall 14 longitudinally spaced from a rear wall 16. A first side wall 18 is laterally spaced from a second side wall 20. The first side wall 18 and the second side wall 20 are coupled to the front wall 14 and the rear wall 16. An intermediate wall 22 extends laterally between the first side wall 18 and the second side wall 20, and is positioned longitudinally between the front wall 14 and the rear wall 16.

[0025] A first or basin bottom wall 24 extends laterally between the first side wall 18 and the second side wall 20, and longitudinally between the front wall 14 and the intermediate wall 22. A second or channel bottom wall 26 extends laterally between the first side wall 18 and the second side wall 20, and longitudinally between the intermediate wall 22 and the rear wall 16. An illustrative drain

channel 28 is defined vertically below the basin bottom wall 24, laterally between the first side wall 18 and the second side wall 20, and longitudinally between the intermediate wall 22 and the rear wall 16.

[0026] With reference to FIGS. 3-5, an arcuate drain wall 32 extends downwardly from the basin bottom wall 24. A drain passageway 34 is defined below the basin bottom wall 24 between the intermediate wall 22 and the arcuate drain wall 32. An opening 36 is defined in the intermediate wall 22 and is in fluid communication between the drain channel 28 and the drain passageway 34.

[0027] The drain assembly 12 is supported by the sink basin 11 and is in fluid communication with the passageway 34 via an opening 40 formed within a lower wall 42. With reference to FIGS. 2-4, the illustrative drain assembly 12 may be of conventional design as including a drain pipe 44 having an upper flange 46 and a tubular body 48 supporting external threads 49. A nut 50 is threadably coupled to the threads 49 of the tubular body 48 to secure to the sink basin 11. A washer 52 (illustratively formed of a polymer) is received intermediate the nut 50 and the flange 46 of the drain pipe 44.

[0028] O-rings 54 (illustratively formed of an elastomer) are received over the drain pipe 44 and provide a seal with the sink basin 11. More particularly, the o-rings 54 provide a seal by tightening the nut 50 with the washer 52, whereby the washer 52 compresses and expands the o-rings 54 to create the seal. An extension pipe 56 includes an internally threaded upper flange 58 and a downwardly extending tubular body 60. The upper flange 58 is coupled to the threads 49 of the drain pipe 44.

[0029] With reference to FIGS. 2-7, an illustrative drain screen or filter 62 is fluidly coupled between the drain channel 28 and the drain passageway 34. The drain screen 62 illustratively includes a resilient body 64 extending axially between a proximal end 66 and a distal end 68. More particularly, the resilient body 64 is received within the opening 36 of the intermediate wall 22 of the sink basin 11. Illustratively, the body 64 of the drain screen 62 is formed of a polymer, such as an elastomer (e.g., a silicone).

[0030] With reference now to FIGS. 8-10, a mounting flange 70 is illustratively supported by the proximal end 66 of the body 64. The mounting flange 70 includes a first side portion 72, a second side portion 74 laterally spaced from the first side portion 72, and an upper portion 76 extending between the first side portion 72 and the second side portion 74. As such, the mounting flange 70 illustratively defines an inverted U shape to assist water flow through a lower portion of the drain screen 62. The mounting flange 70 illustratively engages the intermediate wall 22 to resist the force of water flowing therethrough. A first lower gap 78 is defined by the first side portion 72, and a second lower gap 80 is defined by the second side portion 74 which provides flow passageways to prevent the pooling of water.

[0031] The body 64 further includes a first side wall 82, and a second side wall 84 laterally spaced from the first side wall 82. An upper wall 86 extends between the first side wall 82 and the second side wall 84, and a bottom wall or base 88 extends between the first side wall 82 and the second side wall 84 and is positioned below the upper wall 86. A recess 90 is formed within a lower surface 92 of the base 88. A plurality of axially extending ribs 94 are supported by the base 88 within the recess 90. The recess 90 and the spaced-

apart ribs 94 define channels 95 to facilitate the flow of water through the lower portion of the opening 36 of the intermediate wall 22.

[0032] A strainer 96 is supported by the body 64 distally of the flange 70. The strainer 96 includes a wall 98 having a plurality of openings 100. More particularly, the wall 98 of the strainer 96 extends between the first side wall 82 and the second side wall 84, and includes a center section 102 positioned intermediate a first side section 104 and a second side section 106. The center section 102 illustratively extends proximally of the first side section 104 and the second side section 106. In an illustrative embodiment, the center section 102 of the wall 98 is curved axially towards the proximal end 66 of the body 64. The center section 102 may be of a wide variety of shapes (e.g., U-shaped, V-shaped, etc.). The center section 102 causes debris to move laterally outward toward the side walls 82 and 84 of the body 64 as water flows axially through the strainer 96 from the proximal end 66 toward the distal end 68. Illustratively, the plurality of openings 100 of the strainer 96 are horizontally spaced in a plurality of rows 108, and vertically spaced in a plurality of columns 110.

[0033] With reference now to FIGS. 11 and 12, a further illustrative drain screen or filter 62' is shown. The drain screen 62' includes many similar features as the drain screen 62. As such, in the following description similar components are identified with like reference numbers.

[0034] The drain screen 62' illustratively includes a resilient body 64' received within the opening 36 of the intermediate wall 22 of the sink basin 11. Illustratively, the body 64' of the drain screen 62' is formed of a polymer, such as an elastomer (e.g., a silicone).

[0035] A recess 90 is formed within lower surface 92 of the base 88 of the body 64'. An axially extending rib 94' is supported by the base 88 of the body 64' and extends in a direction from the proximal end 66 toward the distal end 68. The axially extending rib 94' is supported by the base 88 within the recess 90 and defines channels 95' to facilitate the flow of water through the lower portion of the opening 36 of the intermediate wall 22.

[0036] A strainer 96' is supported by the body 64' distally of the flange 70. The strainer 96' includes a wall 98' having a plurality of openings 100. The wall 98' extends between the first side wall 82 and the second side wall 84, and includes a center section 102' positioned between a first side section 104' and a second side section 106'. The first side section 104' and the second side section 106' illustratively include planar walls 105 and 107, respectively, including openings 100.

[0037] The center section 102' includes a curved center wall 112 extending between opposing side walls 114 and 116. A plurality of ribs 118 are supported by the body 64' to frictionally engage with the sink basin 11 when the drain screen 62' is positioned within the opening 36.

[0038] The side walls 114 and 116 are illustratively planar and extend axially from the center wall 112 toward the distal end 68. The side walls 114 and 116 provide support and stability by extending from the center section 102'. More particularly, the side walls 114 and 116 resist the collapsing or compressing of the wall 98'. The side walls 114 and 116 also provide opposing gripping surfaces for a user to facilitate removal of the drain screen 62' from the opening 36 of the sink basin 11 for subsequent cleaning.

[0039] Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the spirit and scope of the invention as described and defined in the following claims.

1. A drain screen for a hidden drain sink, the drain screen comprising:

- a resilient body extending axially between a proximal end and a distal end;
- a mounting flange supported by the proximal end of the body;
- a strainer supported by the body distally of the flange; and wherein the strainer includes a wall including a plurality of openings, the wall being curved axially toward the proximal end of the body.

2. The drain screen of claim 1, wherein the body is formed of a polymer.

3. The drain screen of claim 2, wherein the body is formed of a silicone.

4. The drain screen of claim 1, wherein the wall of the strainer includes a curved section positioned intermediate a first side section and a second side section.

5. The drain screen of claim 1, wherein the mounting flange includes:

- a first side portion;
- a second side portion laterally spaced from the first side portion;
- an upper portion extending between the first side portion and the second side portion;
- a first lower gap defined by the first side portion; and
- a second lower gap defined by the second side portion.

6. The drain screen of claim 1, wherein the body includes:

- a first side wall;
- a second side wall laterally spaced from the first side wall;
- an upper wall extending between the first side wall and the second side wall;
- a bottom wall extending between the first side wall and the second side wall, and positioned below the upper wall;

wherein the wall of the strainer extends between the first side wall and the second side wall; and

wherein a recess is formed within a lower surface of the bottom wall.

7. The drain screen of claim 6, further comprising a plurality of axially extending ribs supported by the bottom wall within the recess.

8. The drain screen of claim 1, wherein the plurality of openings of the strainer are horizontally spaced in a plurality of rows, and vertically spaced in a plurality of columns.

9. A drain screen for a hidden drain sink, the drain screen comprising:

- a resilient body extending axially between a proximal end and a distal end;
- a mounting flange supported by the proximal end of the body;
- a strainer supported by the body distally of the flange; and wherein the strainer includes a wall including a plurality of openings, the wall including a center section positioned intermediate a first side section and a second side section, the center section extending proximally of the first side section and the second side section.

10. The drain screen of claim 9, wherein the body is formed of a polymer.

11. The drain screen of claim 10, wherein the body is formed of a silicone.

12. The drain screen of claim 9, wherein the center section of the strainer is curved toward the mounting flange.

13. The drain screen of claim 9, wherein the body includes:

- a first side wall;
- a second side wall laterally spaced from the first side wall;
- an upper wall extending between the first side wall and the second side wall;
- a bottom wall extending between the first side wall and the second side wall, and positioned below the upper wall;

wherein the wall of the strainer extends between the first side wall and the second side wall; and

wherein a recess is formed within a lower surface of the bottom wall.

14. The drain screen of claim 13, further comprising a plurality of axially extending ribs supported by the bottom wall within the recess.

15. A hidden drain sink comprising:

a basin including:

- a first side wall,
- a second side wall laterally spaced from the first side wall,
- a front wall extending between the first side wall and the second side wall,
- a rear wall extending between the first side wall and the second side wall, the rear wall longitudinally spaced from the front wall,

an intermediate wall positioned between the front wall and the rear wall, a drain opening defined within the intermediate wall,

a drain channel extending between the intermediate wall and the rear wall,

a bottom wall extending between the first side wall and the second side wall; and

a drain screen including:

- a resilient body extending axially between a proximal end and a distal end, the resilient body received within the opening of the intermediate wall of the basin,

a strainer supported by the body distally of the flange, and

wherein the strainer includes a wall having a plurality of openings.

16. The hidden drain sink of claim 15, wherein the wall of the strainer is curved axially toward the proximal end of the body.

17. The hidden drain sink of claim 15, wherein the drain screen further includes a mounting flange supported by the proximal end of the body.

18. The hidden drain sink of claim 15, wherein the wall of the strainer includes a curved section positioned intermediate a first side section and a second side section.

19. The hidden drain sink of claim 15, wherein the body of the drain screen includes:

- a first side wall;
- a second side wall laterally spaced from the first side wall;
- an upper wall extending between the first side wall and the second side wall;

a bottom wall extending between the first side wall and the second side wall, and positioned below the upper wall;

wherein the wall of the strainer extends between the first side wall and the second side wall; and

wherein a recess is formed within a lower surface of the bottom wall.

20. The hidden drain sink of claim **19**, further comprising a plurality of axially extending ribs supported by the bottom wall within the recess of the drain screen.

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